CX-400 Belt Conveyor Drive

Operation & Maintenance Manual

Model CX-400-MD · 4 kW drive motor · 600 mm belt · Document OM-CX400 rev. C

Contents

Section	Page
1. Safety	2
2. Technical data	3
3. Monitoring parameters and alarm limits	4
4. Troubleshooting	5
5. Maintenance schedule	6

Read this manual before installing, operating or servicing the conveyor. Keep it next to the machine.

1. Safety

Warning. Isolate and lock out the drive at the main disconnect before touching rotating parts, the belt, pulleys or the motor housing. Verify zero energy before work starts.

1.1 Qualified personnel

Only qualified maintenance personnel may open the motor terminal box or the gearbox. Operators may clear jams only with the conveyor stopped and locked out.

1.2 Hot surfaces

The motor frame and the drive-end bearing housing can exceed 80 °C in normal operation after a long run. Allow to cool or wear heat-resistant gloves.

1.3 Emergency stop

Pull-cord emergency stops run the full length of the conveyor. After an emergency stop, find and remove the cause before resetting.

2. Technical data

Item	Value
Drive motor	4 kW, 415 V, 3-phase, 1450 rpm, IE3
Rated motor current	8.5 A
Belt width / length	600 mm / 18 m
Nominal belt speed	1.2 m/s
Maximum distributed load	110 kg on the belt
Bearings	Drive end 6308-2RS, non-drive end 6306-2RS
Lubrication	Lithium complex grease, NLGI 2
Ambient temperature	0 to 40 °C

The conveyor is normally fed by an upstream feeder hopper and discharges to an inspection station. Feed rate must not exceed the rated load of the belt.

3. Monitoring parameters and alarm limits

Connect the sensors below to the plant monitoring system using the monitoring tag names shown. Sample each parameter once per second (1 Hz).

Monitoring tag	Description	Unit	Normal range	Warning	Critical
`belt_speed`	Belt linear speed	m/s	0.8 – 1.5	below 0.6 or above 1.7	below 0.3 or above 2.0
`motor_temp`	Drive motor frame temperature	°C	40 – 75	above 80	above 90
`motor_current`	Drive motor current, average of three phases	A	6 – 10	above 12	above 14
`vibration_rms`	Drive-end bearing vibration velocity, RMS	mm/s	1.5 – 4.5	above 5.0	above 7.1
`load`	Load on the belt (load cells)	kg	0 – 80	above 95	above 110

Vibration limits follow ISO 10816-3 zone boundaries for a machine of this size on a rigid foundation. A warning means plan an inspection; a critical value means stop the conveyor as soon as it is safe to do so.

4. Troubleshooting

Symptom	Likely cause	Check first
Vibration rises first, then motor and bearing temperature rise	Drive-end bearing wear or lubrication failure	Measure the drive-end bearing housing with an IR thermometer and listen for grinding. Above 90 °C stop the conveyor and replace the bearing.
Motor temperature rises while current and vibration stay normal	Cooling fan failure or blocked fan cowl	Check that the fan turns and the cowl and fins are clear of dust and product.
Belt speed drops while motor current falls	Belt slip, low belt tension or a glazed drive drum	Re-tension the belt to specification and inspect the drum lagging.
Load and motor current rise, belt slows	Jam or overload, often from upstream over-feed	Stop and lock out, clear the jam, then compare the feeder rate with the rated load.
A reading drops to zero or stays perfectly flat	Sensor, cable or gateway fault	Compare with a handheld instrument and check the sensor cable and connector.

5. Maintenance schedule

Interval	Task
Daily	Walk the line, listen for unusual noise, check belt tracking and the emergency pull-cords.
Every 500 h	Check belt tension and the drive drum lagging. Clean the motor fan cowl.
Every 2000 h	Re-grease the drive-end and non-drive-end bearings with NLGI 2 grease, 8 g per bearing.
Every 8000 h	Replace both bearings. Check pulley alignment and motor mounting bolts.

End of the sample manual. This document describes fictional equipment and exists only to demonstrate BatchGuard AI's manual extraction.